Polynomials – 7

Interpolation and finite differences

Homework

8 сентября 2026 года

Theoretical minimum. Lagrange's formula and the Vandermonde determinant; the difference operator Δ and the constancy of the n-th difference; the alternating sum identity; Newton's form with binomial coefficients; integer-valued polynomials; divided differences for unequal nodes.

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1. Prove that a polynomial taking integer values at all integers is an integer combination of the binomial coefficients $\binom{x}{k}$, and give an example of such a polynomial with non-integer coefficients of every degree.

2. (Putnam 2007) Let k be a positive integer. Prove that there are polynomials $P_0(n), \dots, P_{k-1}(n)$, possibly depending on k, such that for every integer n $\left\lfloor \frac{n}{k} \right\rfloor^k = P_0(n) + P_1(n)\left\lfloor \frac{n}{k} \right\rfloor + \cdots + P_{k-1}(n)\left\lfloor \frac{n}{k} \right\rfloor^{k-1}$.

3. (Высшая лига 2012) Define polynomials by $g_0(x) = 1$ and $g_{n+1}(x) = (x + x^{-1})g_n(x)$ with the term containing x^{-1} omitted; for instance $g_1 = x$ and $g_2 = 1 + x^2$. Find $g_{15}(0)$ and $g_{16}(0)$.

4. Let f be the polynomial of degree n with $f(k) = 2^k$ for $k = 0, 1, \dots, n$. Compute $f(n+1)$ and $f(n+2)$, and explain the answer through the finite differences of the sequence 2^k.

5. (Karasev 5.12) Let P be a polynomial of degree n with integer coefficients, not identically zero. Prove that $|P(x)| \geqslant e^{-n}$ for some $x \in [0, 1]$.

6. Prove Newton's forward difference formula and use it to evaluate $\sum_{k=1}^{m} k^3$ and $\sum_{k=1}^{m} k(k+1)(k+2)$ without induction.

7. Let $P \in \mathbb{Z}[x]$ take the value 7 at four distinct integers. Prove that $P(k) \neq 14$ for every integer k, and find the general statement behind this.

8. Prove the Vandermonde determinant formula and deduce that the interpolating polynomial through $n+1$ nodes in $[0,1]$ with pairwise distances at least $\delta > 0$ has coefficients bounded by a constant depending only on n and δ times the maximum of the values.

9. Prove that if $\deg P \leqslant n$ and $|P(k)| \leqslant 1$ for $k = 0, 1, \ldots, n$, then $|P(x)| \leqslant 2^n$ for all $x \in [0, n]$, and show that the exponential growth cannot be avoided.

10. Prove the divided difference formula $P[x_0, \ldots, x_k] = \sum_i \frac{P(x_i)}{\prod_{j \neq i}(x_i - x_j)}$ and Newton's interpolation formula for unequal nodes; deduce that the divided difference of order n of a polynomial of degree n equals its leading coefficient.